Poojan Patel

Graduate Software Developer

485 Visa | Full Working Rights

Sydney, NSW | 0422 000 830 | patelpoojan6225@gmail.com | Portfolio | GitHub | LinkedIn

SUMMARY

Results-driven IT graduate with hands-on experience building scalable backend systems and full-stack applications. Strong foundation in Python, Java, Go, SQL, and distributed systems, with practical exposure to API design, database optimisation, and Agile development. Passionate about delivering reliable, secure, and maintainable software solutions.

SKILLS

Languages: Java, Python, Go, JavaScript, TypeScript, Dart, SQL, C#, C, C++, HTML, CSS
Frameworks & Tools: React, Django, Flutter, Firebase, MongoDB, MySQL, PostgreSQL, Git, Chi, Supabase
Concepts: OOP, Data Structures & Algorithms, Distributed Systems, API Design, Software Architecture, Data Modelling
Others: Unity, Node.js, CI/CD (GitHub Actions), PHPUnit, Playwright
Generative AI: Anthropic Claude API, LLM integration (server-side), prompt engineering, AI-assisted grading pipelines

EXPERIENCE

Robotics Marketer: Melbourne, VIC — *Software Developer Intern*
JULY 2025 – NOVEMBER 2025 **Tech:** Django, MongoDB, MongoEngine, Python, JavaScript

- Delivered backend features for a production web application used by 4000+ users, improving API response efficiency by 22%.
- Designed and implemented RESTful APIs supporting frontend data consumption and asynchronous operations.
- Modelled and optimised NoSQL schemas, improving data consistency and scalability.
- Collaborated with a team of 7 in an Agile environment; participated in code reviews, testing, and deployment preparation; produced clear API and database documentation.

Chemist Warehouse: Sydney, NSW — *Casual Worker*
APRIL 2023 – PRESENT

- Operated across all major warehouse functions in a fast-paced environment, demonstrating adaptability and strong attention to detail.
- Collaborated with team members to meet daily productivity targets and tight deadlines.

Taral Sir's IELTS: Ahmedabad, India — *Tutor*
AUGUST 2022 – SEPTEMBER 2022

- Taught IELTS preparation to classes of 70–80 students; reviewed 250+ test papers weekly and provided structured feedback.

PROJECTS

Selective Prep — *Full Stack Web Application*
Tech: React, TypeScript, Go, PostgreSQL, Anthropic Claude API

- Built a full-stack exam practice platform for the NSW Selective Placement Test — 9 practice tests, 8 question types, and 3 exam sections — with a Go/Chi REST API, PostgreSQL via Supabase, and a React/TypeScript frontend.
- Designed a JSONB-backed schema for flexible multi-format test content with indexed queries; implemented full attempt history, session tracking, exam/practice timer modes, and a protected admin panel.
- Integrated Claude Haiku server-side to AI-mark student essays against 6 NSW marking criteria, keeping the API key entirely server-side; built Recharts analytics dashboards for per-student score history and skill breakdowns.
- Deployed frontend on Vercel and backend on Railway with auto-deploy on push; configured CORS, admin secret middleware, and environment-based API key management.

Prime Towing & Hauling — *Full Stack Business Website*
Tech: HTML, CSS, JavaScript, PHP, MySQL, PHPMailer, PHPUnit, Playwright, GitHub Actions

- Built and deployed a production website for a real Sydney towing business on a live Hostinger domain; engineered a PHP/MySQL backend with async contact form handling, a visitor analytics tracker (device, browser, IP hash, referrer), and a brute-force-protected admin dashboard with DB-backed IP throttling.
- Developed a responsive frontend with vanilla JavaScript: infinite-loop gallery carousel, parallax hero banner, scroll-reveal animations via IntersectionObserver, and a mobile hamburger menu.

- Established a CI/CD pipeline via GitHub Actions — PHPUnit and Playwright tests run against a real MySQL service container, followed by automated image compression and FTP deployment to production.

BargainHunter — *Full Stack Mobile Application*

Tech: Flutter, Dart, Firebase (Auth, Firestore, Storage), Provider

- Built a cross-platform community deal-sharing app (Android, iOS, Web) with real-time Firestore streaming, Firebase Auth, and Cloud Storage image uploads with UUID-based naming.
- Implemented multi-criteria client-side filtering (keyword, price range, store) using Provider/ChangeNotifier; integrated image_picker with validation before upload; built a deal expiry countdown parsing ISO8601 timestamps with visual indicators for expired deals.

Cataclysm — *Interactive Game Development*

Tech: Unity, C#, Blender

- Developed a 3D space exploration game showcased at Macquarie University Astronomy Night for 2 consecutive years; built an asteroid cluster of ~1,000 randomised asteroids across 4 models, mouse-based joystick controls, a 10–12s trail effect for creative drawing, and a 3-minute session timer to manage queue flow.
- Led all C# gameplay logic — collision deviation, hidden ship objectives, and sandbox free-roam — coordinating 3D asset creation and sound effects across a 3-person team.

WaterFlo DSL — *Domain-Specific Language / Compiler Design*

Tech: Java, Compiler Design, AST, Lexer / Parser, Language Design

- Designed and implemented a custom programming language in Java to model and simulate river flow dynamics, building the full pipeline from scratch: lexer, recursive-descent parser, AST construction, and tree-walking interpreter.
- Enabled structured execution of domain-specific programs including multi-day dam simulations and flow algorithms, demonstrating deep understanding of language design, evaluation semantics, and compiler theory.

CricketOracle — *Machine Learning / Data Science*

Tech: Python, Scikit-learn, Pandas, Machine Learning

- Built a machine learning model to predict cricket match outcomes using historical data, player statistics, and match conditions; achieved ~60–70% pre-match accuracy with performance improving on additional contextual inputs.
- Applied end-to-end ML pipeline including data wrangling with Pandas, feature engineering, model selection, and evaluation — demonstrating practical data-driven decision making aligned with the Data Science minor.

EDUCATION

Bachelor of Information Technology — Macquarie University, Sydney

FEB 2023 – DEC 2025

Major: Software Technology, Minor: Data Science. Coursework: algorithms, distributed systems, mobile dev, databases, OS, graphics, big data, and cybersecurity.

VOLUNTEERING

Student Representative – COMP1010, Macquarie University | Session 2, 2023

- Acted as liaison between students and unit coordinators, improving communication and providing feedback.

Unity Game Developer – Astronomy Night Showcase, Macquarie University

- Developed and presented Cataclysm to children and families across 2 annual showcases, demonstrating programming concepts in an accessible and engaging way.